Operations Management

Session 7: Supply Chain Dynamics and Strategies

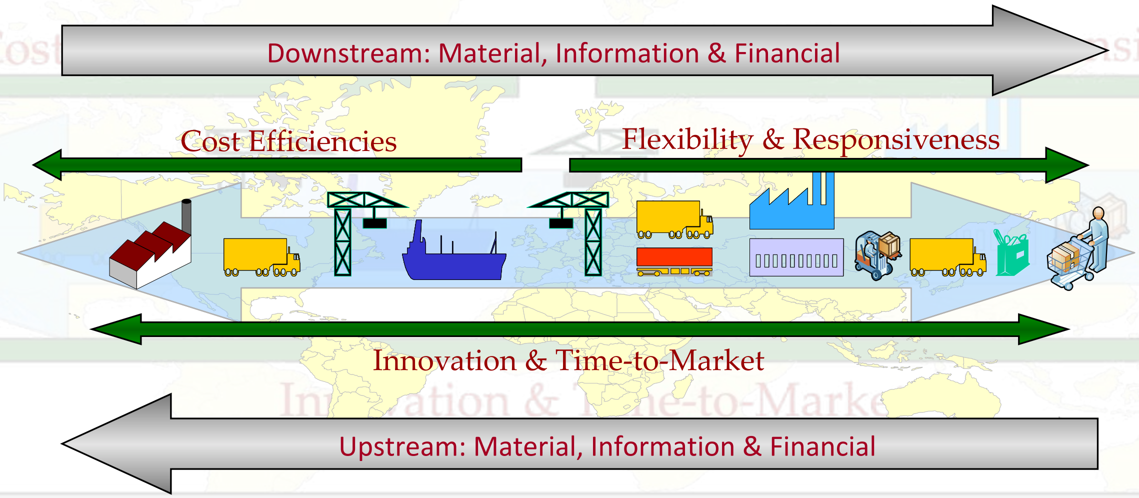
Outline:

- Supply Chain Management & Beer Game (simulation)
  - Play the game (roles, ordering, lead times)
- Debrief
  - What happened to orders/inventory/backlogs?
  - Bullwhip effect: definition + why it occurs
  - Negative implications (cost, service, volatility)
  - What could reduce it? (high-level ideas)
- Strategies for supply chain coordination
  - Information sharing
  - Order smoothing
  - Contracts / incentive alignment(e.g., revenue sharing, buy-back)
  - JITD / VMI

PROF: RENÉ CALDENTEY

Supply Chain Management

"Supply Chain Management is a set of techniques utilized to efficiently **integrate** suppliers, manufacturers, warehouses, and stores, so that merchandise is produced and distributed at the **right** quantities to the **right** locations, and at the **right** time, in order to **minimize** system-wide costs while satisfying **service level** requirements."

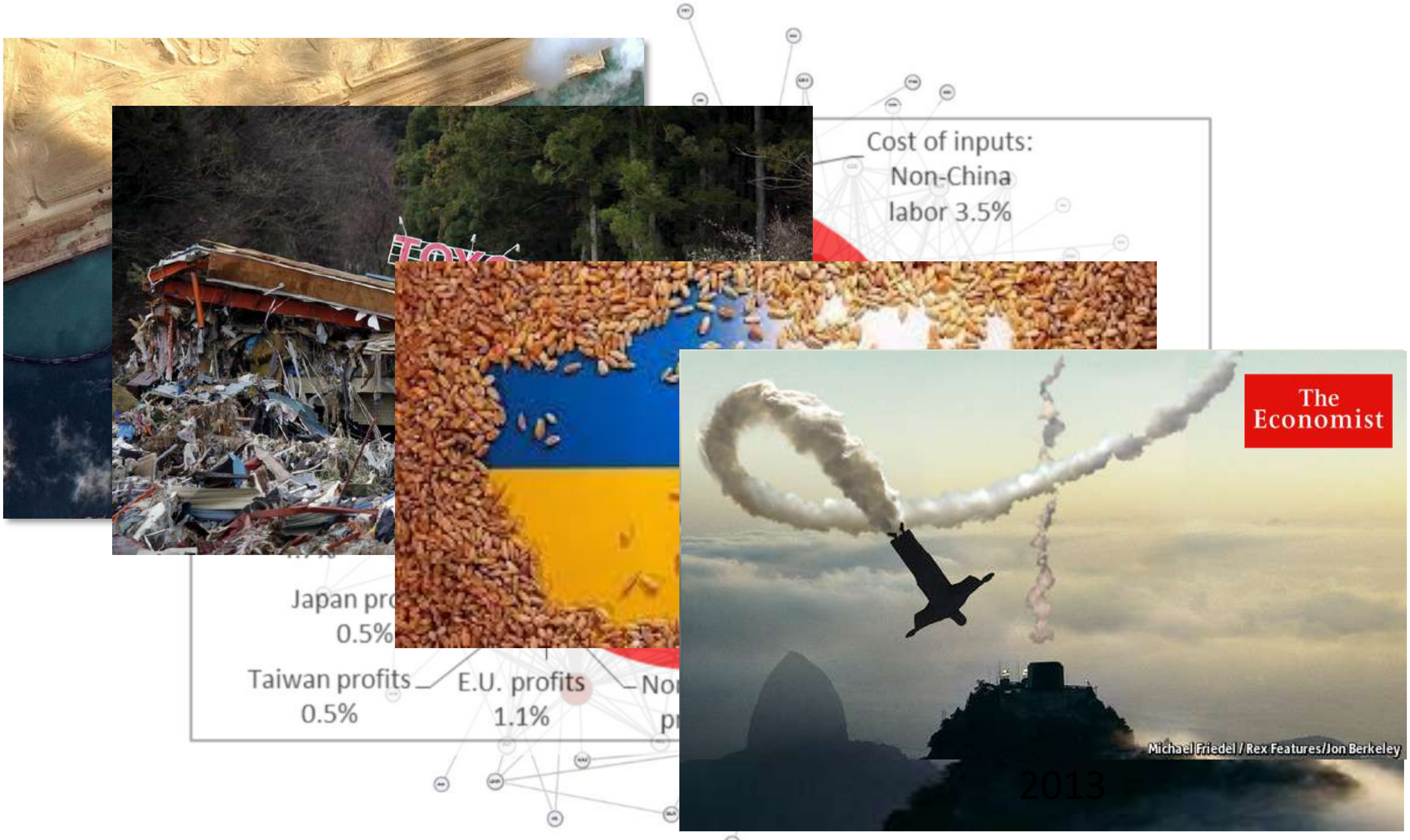


Efficient supply chain should allow for a better competition

- ✓ Products Customization: Dell Computers
- ✓ Quality Improvement: Toyota
- ✓ Cost Reduction: Wal-Mart
- ✓ Speed-to-Market: Intel, Zara

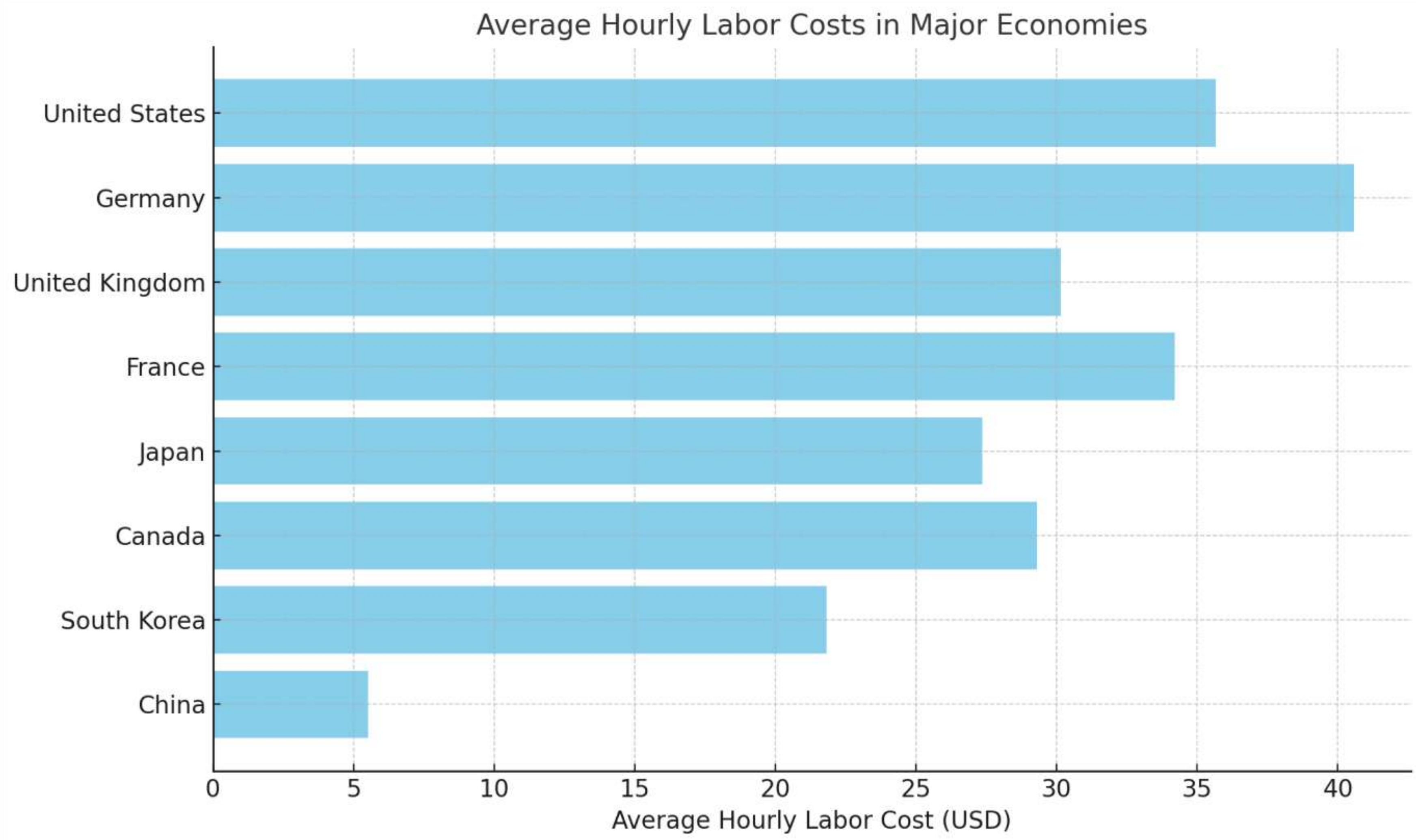
Global Markets

Global Opportunities and Risks



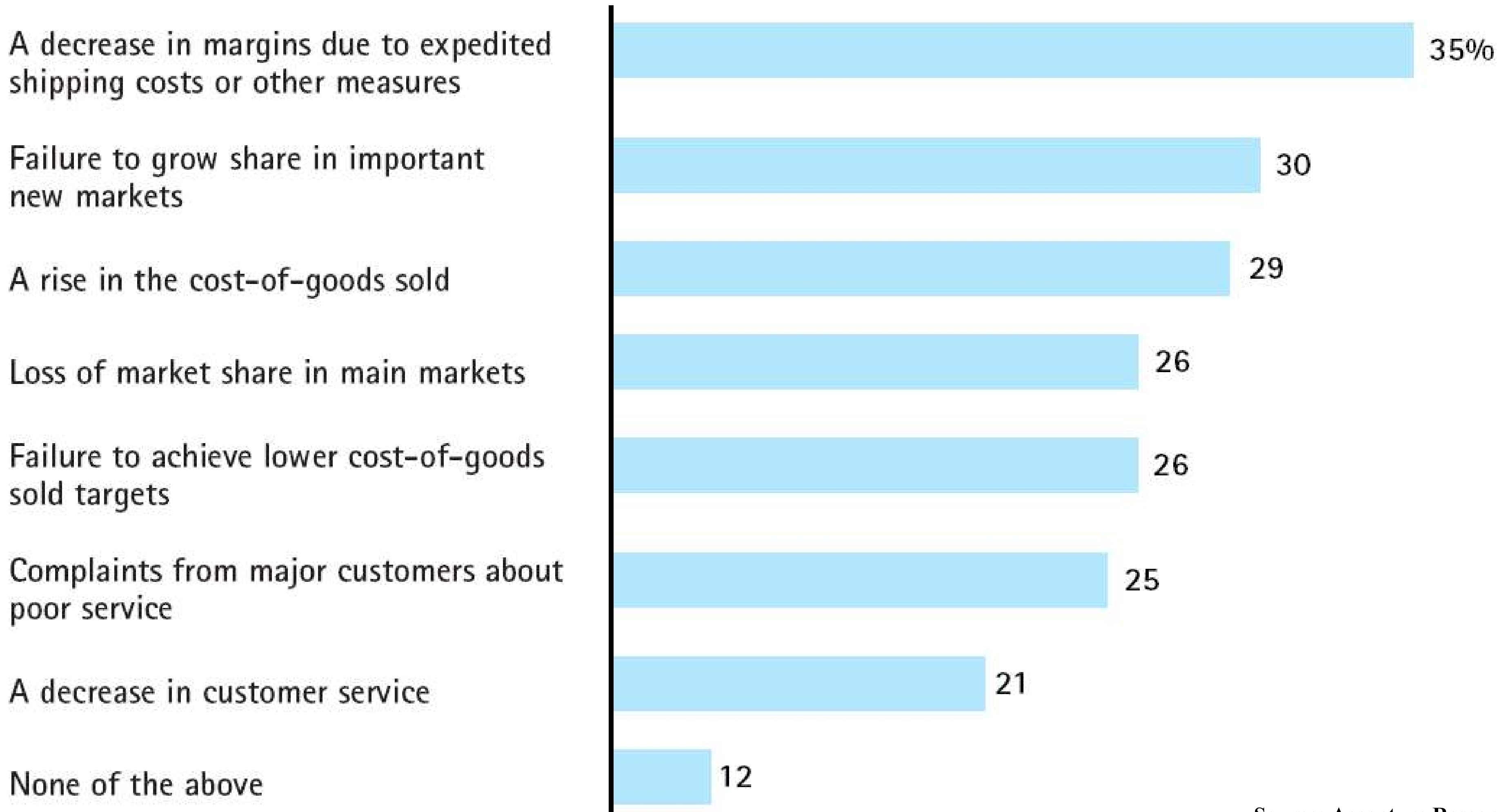
About 70% of international trade today involves global value chains (GVCs), as services, raw materials, parts, and components cross borders – often numerous times.

Can Foxconn still be Profitable?



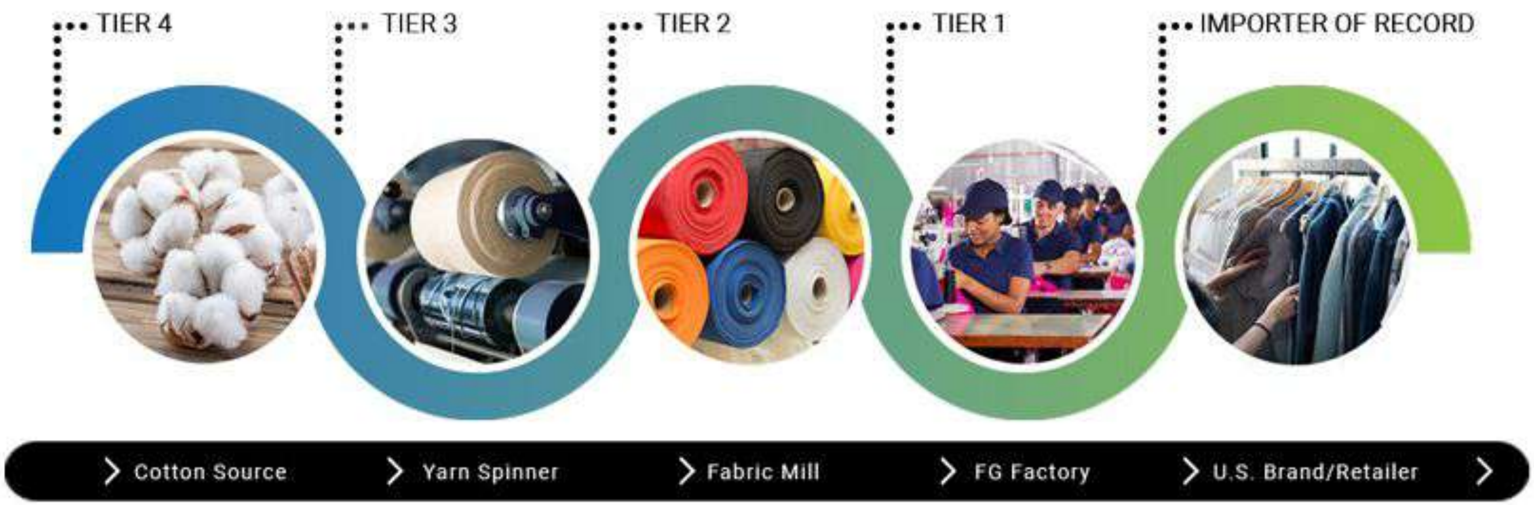
Global SCM Challenges

The cost of poorly designed and executed operations strategies



Source: Accenture Research

Global SCM Challenges
Traceability – Sustainability –Social Responsibility



Top Supply Chains

Gartner Supply Chain Masters for 2018

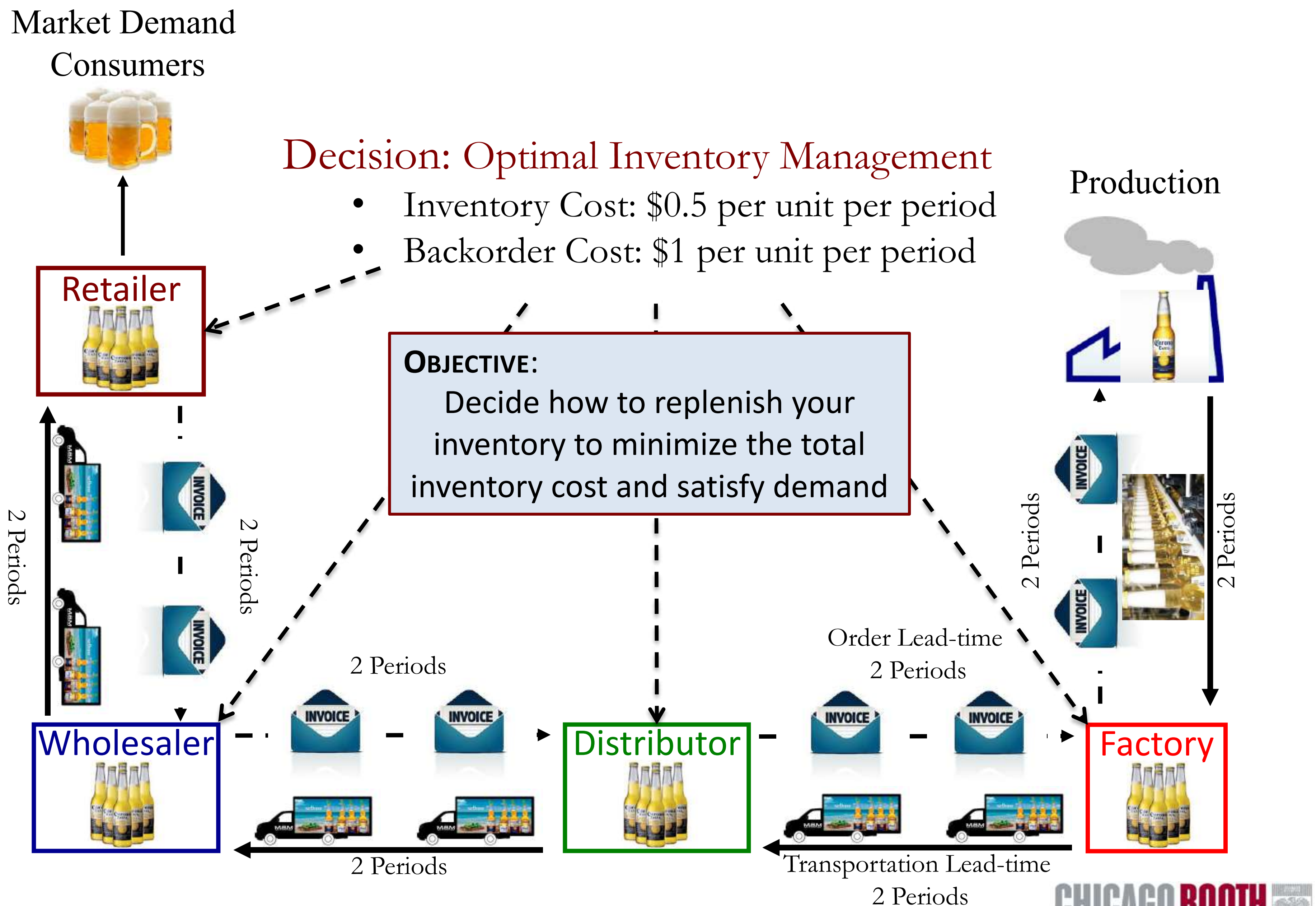
Apple	P&G	Amazon	McDonald's
Years With Top-Five Composite Score	Years With Top-Five Composite Score	Years With Top-Five Composite Score	Years With Top-Five Composite Score
2009 2010 2011 2012 2013 2014 2015 2016 2017 2018	2009 2010 2011 2012 2014 2015 2016 2017 2018	2011 2012 2013 2014 2015 2016 2017 2018	2012 2013 2014 2015 2016 2017 2018

ID: 351344

Rank	Company	Peer Opinion (184 voters) (25%)	Gartner Opinion (42 voters) (25%)	Three-Year Weighted ROA (20%)	Inventory Turns (10%)	Three-Year Weighted Revenue Growth (10%)	CSR Component Score (10%)	Composite Score
1	Unilever	2,413	667	10.3%	7.5	2.6%	10.00	6.36
2	Inditex	1,254	345	16.5%	3.9	10.9%	10.00	4.85
3	Cisco Systems	785	541	7.9%	13.1	-0.4%	10.00	4.41
4	Colgate-Palmolive	898	324	17.6%	5.1	-2.2%	10.00	4.40
5	Intel	831	499	8.9%	3.6	4.8%	10.00	4.36
6	Nike	1,349	270	17.4%	3.8	6.8%	6.00	4.25
7	Nestlé	1,326	426	6.4%	4.8	-0.2%	10.00	4.21
8	PepsiCo	1,094	391	7.3%	8.8	-0,6%	10.00	3.99
9	H&M	760	193	18.1%	2.8	7.8%	10.00	3.96
10	Starbucks	1,040	186	20.4%	11.8	9.2%	4.00	3.85

http://www.gartner.com

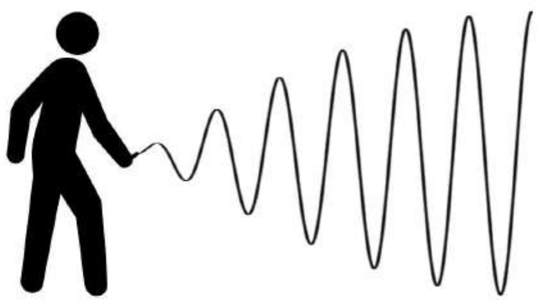
Beer Game Overview

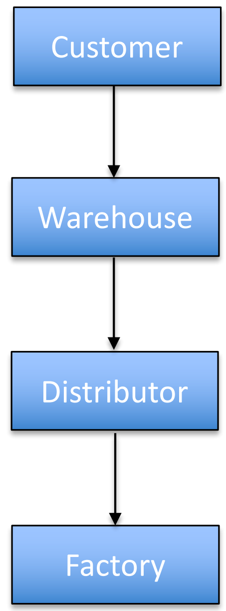


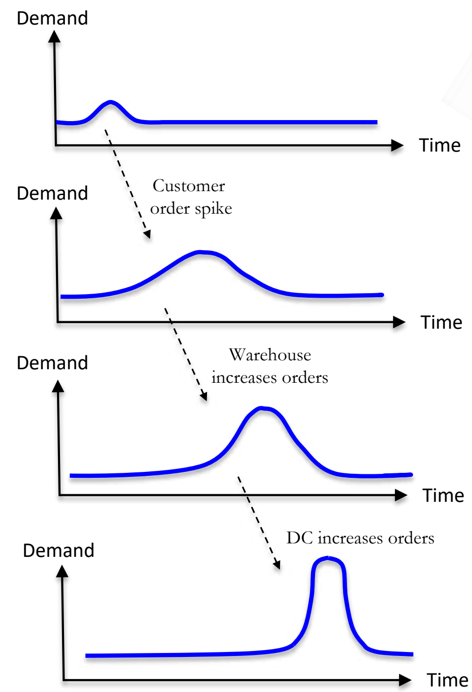
Beer Game Overview

Beer Game Debrief

Information gets distorted for various reasons as it travels upstream in the supply chain. This imposes tremendous costs of oversupply, stockouts, excess capacity investments, and greater risk.







Order volatility leads to:

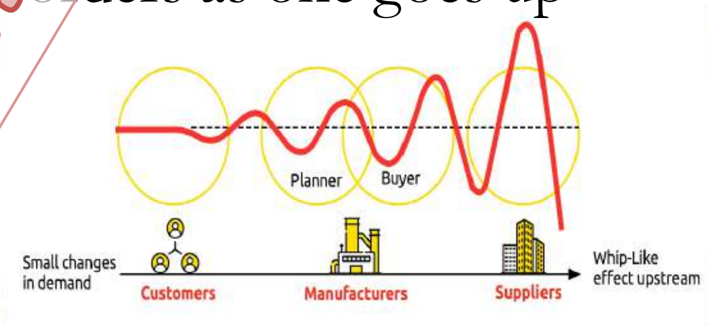
- Increase inventory costs
- Inefficient utilization of capacity
- Excess warehousing

This phenomenon is known as THE BULLWHIP EFFECT.

(Article: The Bullwhip Effect in Supply Chains)

Bullwhip Effect

- **What is it?**
  - Increase in Coefficient of Variation of orders as one goes up the supply chain



- **Where did it come from?**
  - Coined by P&G to describe behavior of diaper orders through the supply chain

- **Industry examples:**
  - GROCERY INDUSTRY: $30 billion of inventory in supply chain, not including the factory and retailer
  - PHARMACEUTICAL INDUSTRY: 100 days of supply scattered throughout the supply chain

How to deal with it?

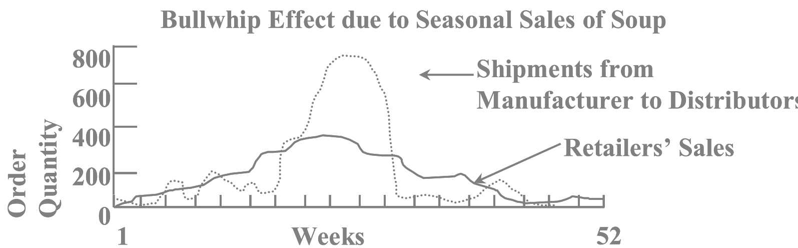
Causes

Demand Forecast Updating

- Managerial overreaction to changes in demand/orders
- Forecasting techniques that rely heavily on recent demand

Order Batching

- Economies of Scale in ordering and transportation
- Periodic planning and ordering as part of MRP



Price Fluctuation

- Quantity Discount
- Trade promotions = "The dumbest marketing ploy ever!"

Rationing and Shortage Gaming

- Demand exceeds manufacturing capacity
- Customers exaggerate their real needs ("gaming")

Supply Chain Strategies

- Information Sharing
- Order Smoothing
- Vendor Managed Inventory
- Incentive Alignment
- Production Postponement
- Distribution & Logistics Management Systems

Information Sharing

The bullwhip effect is amplified when supply-chain decisions are based on distorted demand signals.

- Why information sharing helps:
  - Replaces orders with true demand signals (POS/sell-through)
  - Aligns forecasts across supply-chain tiers
  - Smooths production and replenishment decisions
  - Reduces safety stock driven by demand uncertainty
- What is typically shared in practice:
  - Point-of-sale (POS) data
  - Store and DC inventory levels
  - Inventory in transit
  - Promotion and pricing calendars
- Technology
  - RFID: Companies use RFID tags to track raw materials from suppliers to production lines, reducing errors and delays
  - Blockchain: Provides a shared, tamper-evident ledger that improves cross-firm trust and traceability (e.g., chain-of-custody, certifications, shipment events), reducing reconciliation effort and disputes.

Walmart

CPFR and Luminate Programs

Lufthansa

Lufthansa Technik parts pooling

ZARA

IBM Food Trust™

Order Smoothing

The deliberate stabilization of replenishment orders, prioritizing long-run trends over short-term noise.

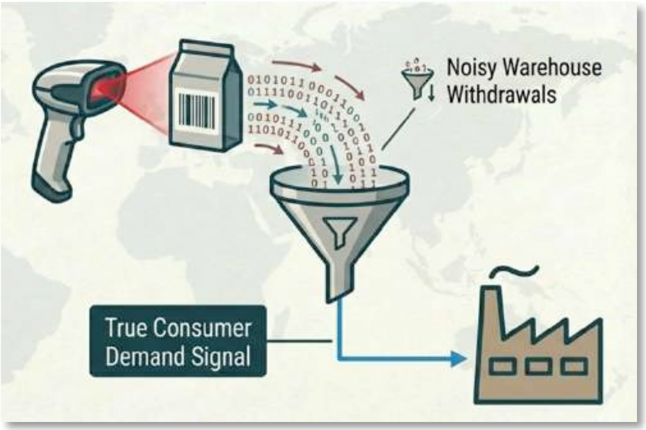
How it works

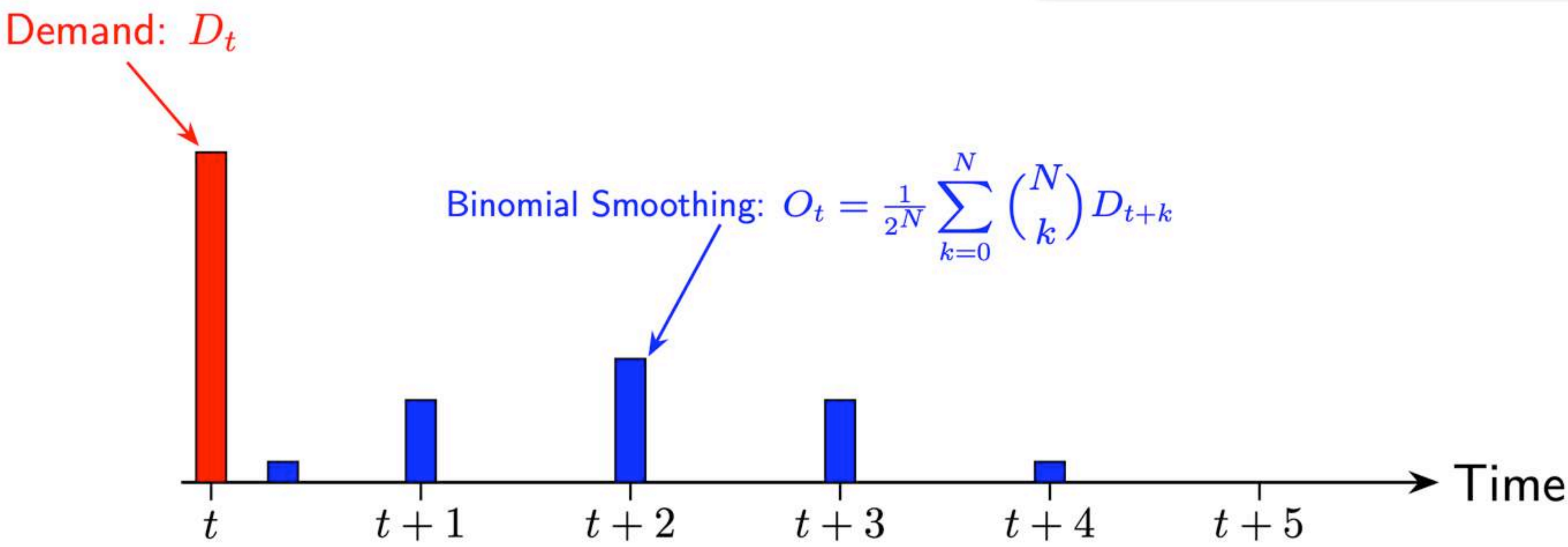
- Start from a **baseline replenishment plan** tied to long-run demand (not last week's sales).
- Use current demand/inventory signals to compute a **target order**, but **adjust toward it gradually** rather than all at once.
- Treat short-term spikes/dips as **noise until confirmed** (avoid overreacting).
- Maintain a steady cadence for production and transportation, using exceptions only when truly needed (promotions, disruptions, stockout risk).

Example rules firms use ("smoothing guardrails"):

- Order change cap: "Weekly order can change by at most ±10–15% vs. last week."
- Partial adjustment: "Close only 30–50% of the gap between current order and target order each cycle."
- Moving-average ordering: "Base orders on 4–8 week average POS (or demand) rather than last week."
- Frozen window: "Next 1–2 weeks are fixed; adjustments allowed only beyond the freeze horizon."
- Min/max + reorder band: "Keep orders within a band unless inventory crosses a trigger threshold."

Delayed Response



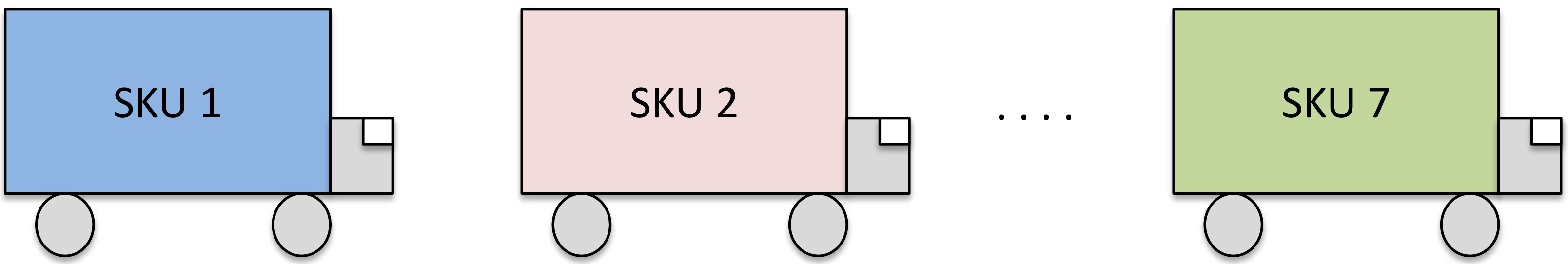


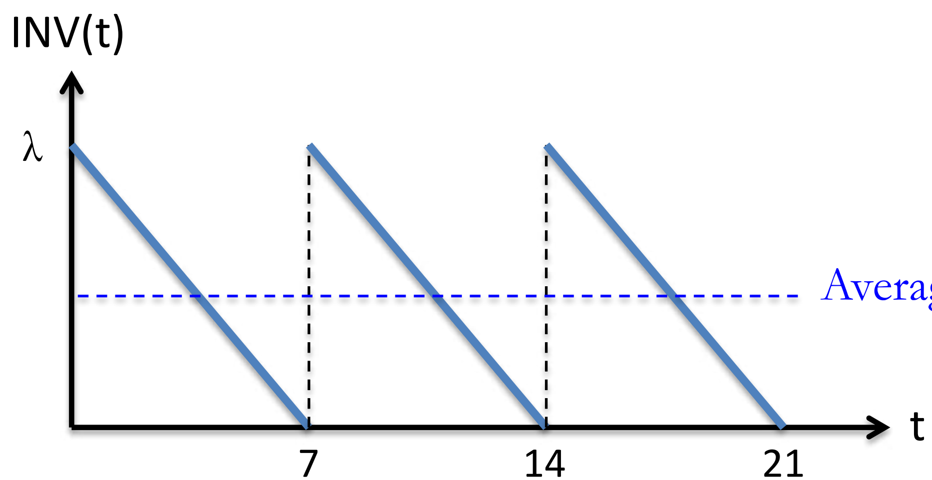
Impact of Smoothing on Logistics

7 SKU's, Weekly demand λ for each SKU

Old System

Daily truckloads each containing one week's supply of one SKU.



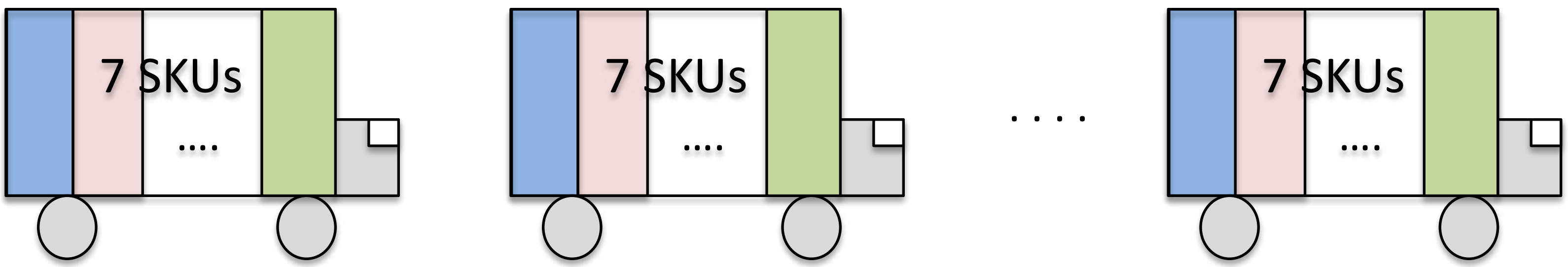


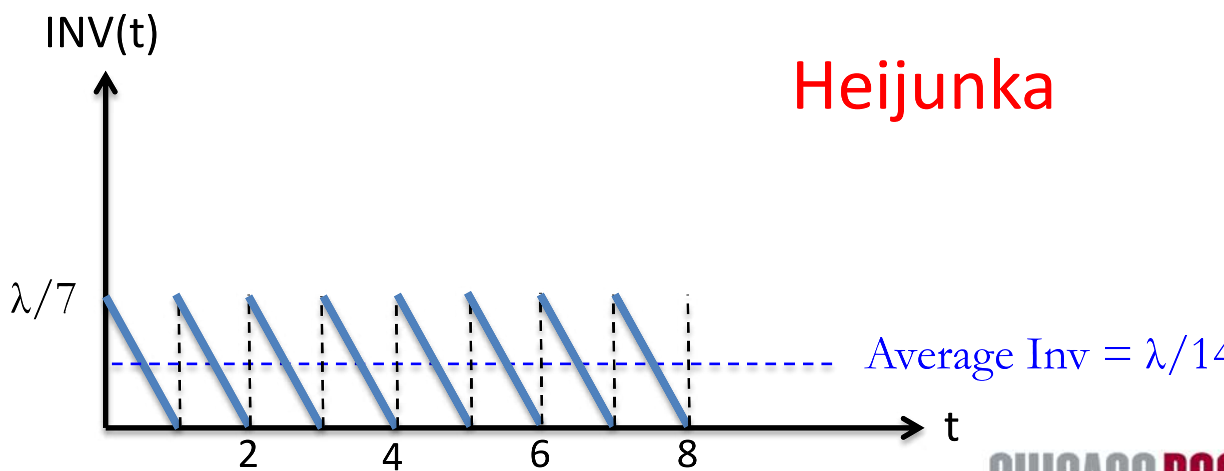
Impact of Smoothing on Logistics (cont'd)

7 SKU's, Weekly demand λ for each SKU

New System

Daily truckloads each containing one day's supply of every SKU.



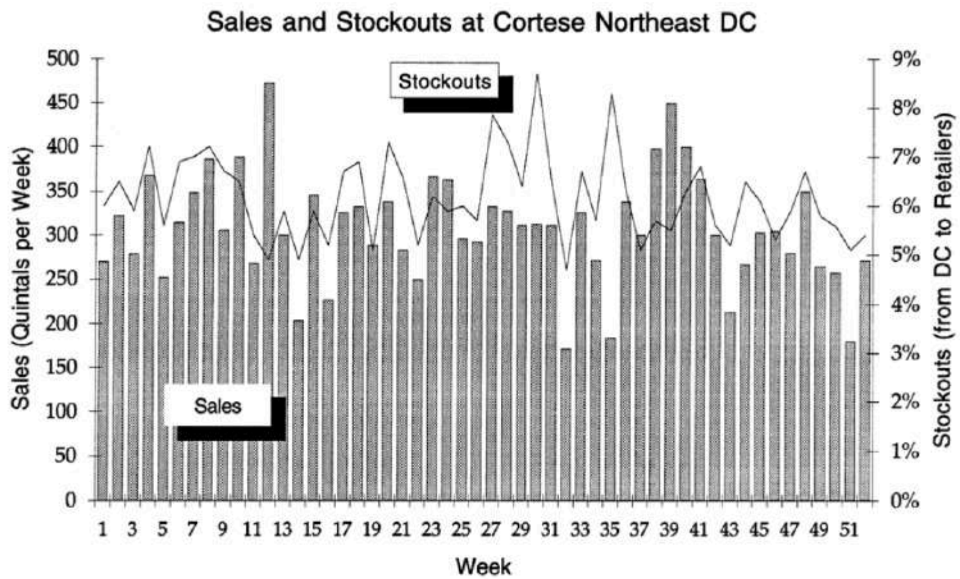


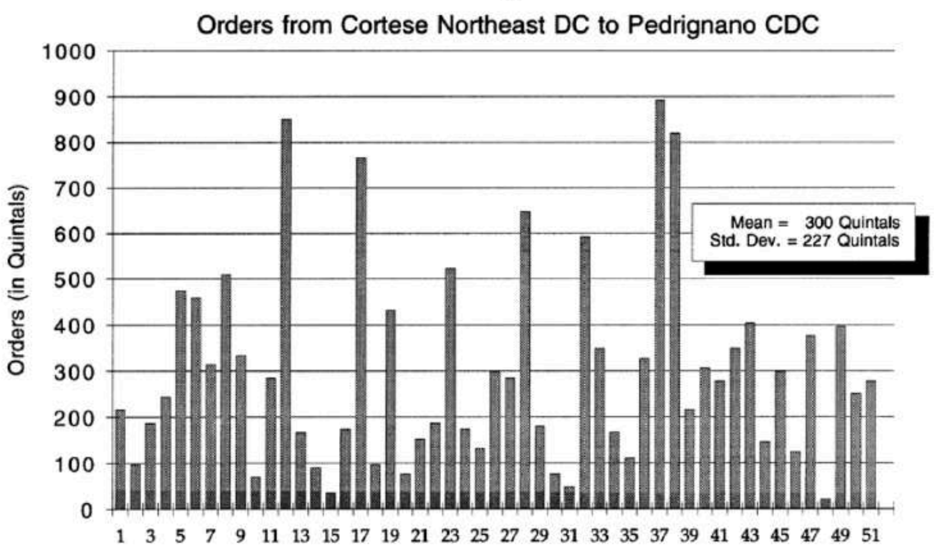
Heijunka

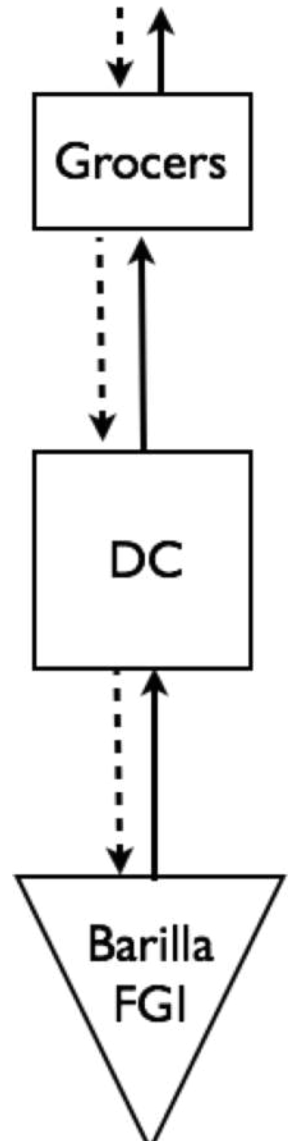
Vendor Managed Inventory

- Key Benefit: Smooths orders → mitigates the bullwhip effect
- Operational Impact: Lower order variance, better capacity utilization

Example: Barilla and the Bullwhip Effect







Causes of Order Variability

- Quantity discounts
  - Transportation discounts: full truckloads (2-3%)
  - Volume discounts: multiple truckloads (4%)
- Salesforce incentives
  - Promotion periods every 4-5 weeks
  - "Sell-in" versus "sell-thru"
- Forecast error: over and under-correction
  - Long and variable lead-times
  - Product proliferation
- Retailer stockouts

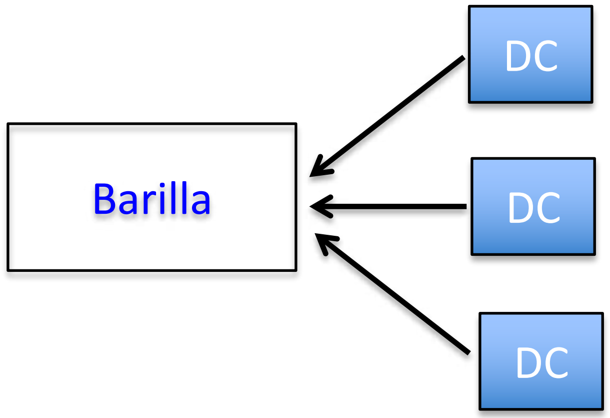
Solution… Vendor Managed Inventory (VMI)

- Distributors give Barilla daily inventory and shipment data by SKU.
- Barilla controls shipping decisions.
  - Also known as Just-in-Time Distribution (JITD).

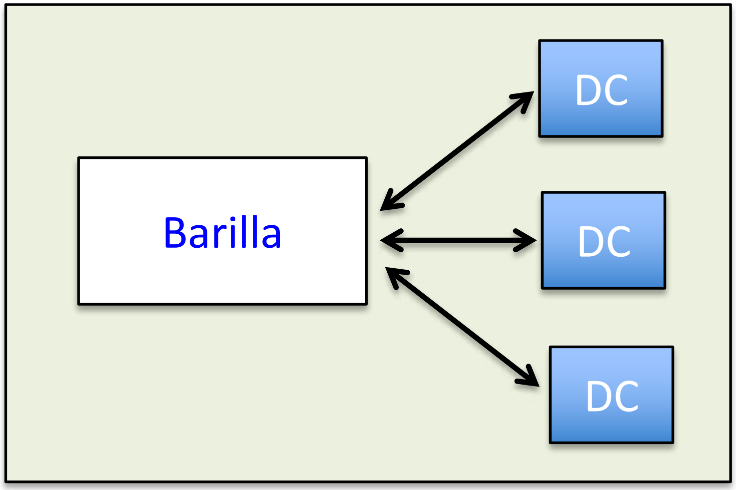
Why would JITD solve their problems?

"What make you thing that you could manage my inventories any better than I can?"

Sequential, Myopic Optimization

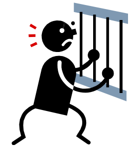


Global Optimization

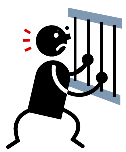


Supply Chain Coordination

Prisoner's Dilema



	Collaborate	Not Collaborate
Collaborate	(5,5)	(1,10)
Not Collaborate	(10,1)	(2,2)



Double Marginalization: Each player maximizes their profit with respect their own margin, not that of the chain.

Decentralized Supply Chain

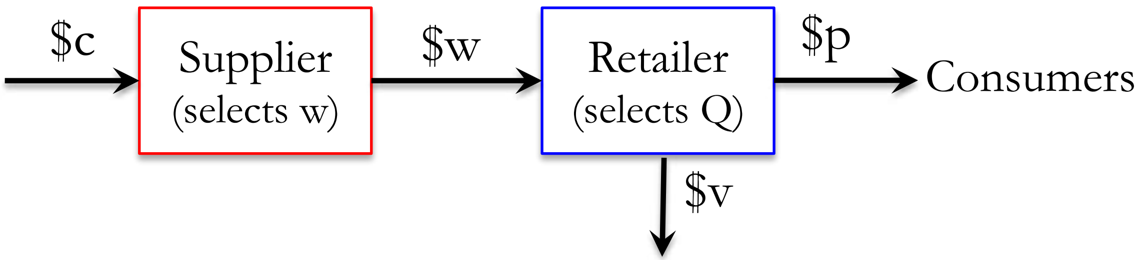
$c → Supplier → $w → Retailer → $p

Integrated Supply Chain

$c → Chain → $p

Contracting with the newsvendor

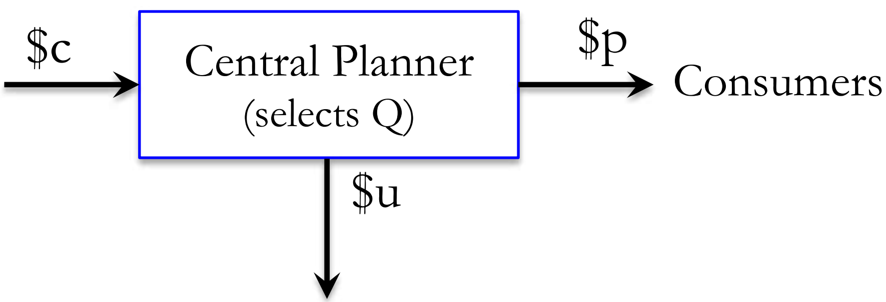
Decentralized Supply Chain



Retailer's Optimal Inventory

$$\mathrm{Prob}(D \leq Q^R) = \frac{p - w}{p - v} = SL^R$$

Integrated Supply Chain



Supply Chain Optimal Inventory

$$\mathrm{Prob}(D \leq Q^C) = \frac{p - c}{p - u} = SL^C$$

Typically, economics incentives are misaligned $SL^R \leq XL^C$.

Questions:

- In a standard wholesale price contract, who absorbs demand risk?
- Who is in a better position to handle excess inventory?
- How to modify the economics incentives to implement optimal inventory decisions?

Double Marginalization

Decentralized Supply Chain

$$\xrightarrow{\$c} \boxed{\text{Supplier (selects w)}} \xrightarrow{\$w} \boxed{\text{Retailer (selects Q)}} \xrightarrow{\$p} \text{Consumers}$$

Retailer $\downarrow$ $\$v$

Retailer's Optimal Inventory

$$\text{Prob}(D \leq Q^R) = \frac{p - w}{p - v} = SL^R$$

Example: Tech Fiber (TF) produces jacket and sells to Ski Adventure (SA) which sells them in the market. Unsold jackets have no salvage value. How many jackets should SA procure?

$$\xrightarrow{\text{Cost } c = \$10} \boxed{\text{TF}} \xrightarrow{\text{Wholesale Price } w = \$100} \boxed{\text{SA}} \xrightarrow{\text{Market Price } p = \$200}$$

Market Demand
$D \sim N(1000, 300^2)$

Retailer's Newsvendor Solution:

Critical Fractile $= \frac{c_s^R}{c_s^R + c_e^R} = 0.5 \quad \Longrightarrow \quad Z_{0.5} = 0 \quad \Longrightarrow \quad Q^R = \mu + \sigma \times Z_{0.5} = 1000$ units

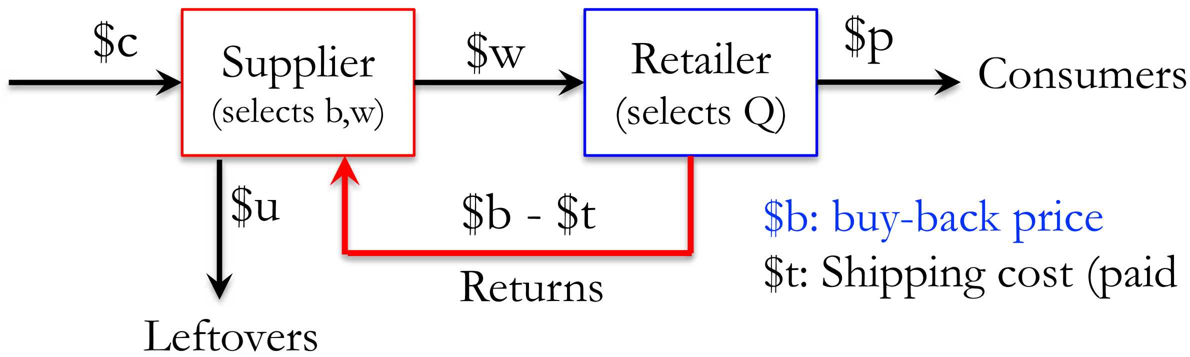
Supplier's Profits: $Q^R \times (\$100 - \$10) = \$90{,}000$

Retailer's Expected Profit: $(Cs+Ce) \times E[\underbrace{\text{Min}\{D, Q^R\}}_{\mu - \sigma \cdot L(Z_{0.5}) = 880.3}] - Ce \times Q^R = \$76{,}063$

Inventory Management Agreements

Buy-back Contracts

A business arrangement whereby one party sells inventory to a second party, with the promise to repurchase the inventory at a future point in time.



$b: buy-back price

$t: Shipping cost (paid by retailer)

Inventory Consignment

The inventory is stored at the buyer's location, but the supplier only charges the buyer once the goods are sold or consumed.

Third-party logistics (3PL) Inventory System (e.g., FBA)

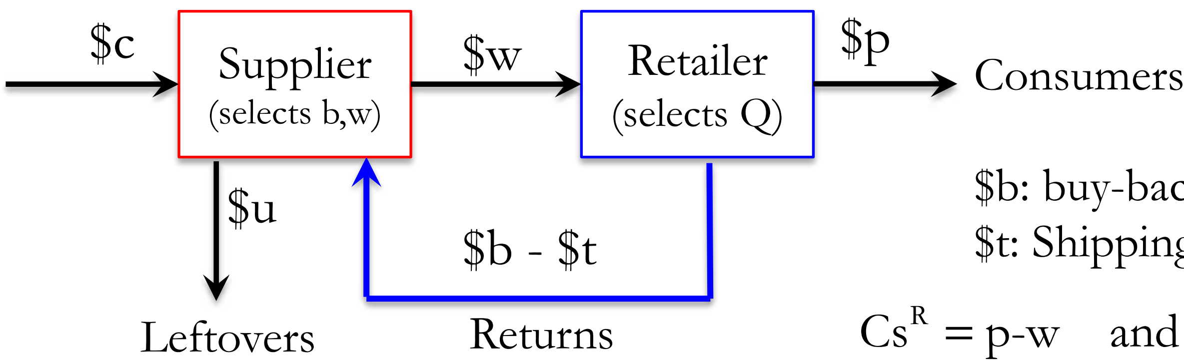
Sellers retain ownership of their inventory while 3PL (e.g., Amazon) provides storage, packaging, shipping, returns and customer service.

Buy-back contracts

Definition (Buy-back agreement)

A business arrangement whereby one party sells inventory to a second party, with the promise to repurchase the inventory at a future point in time.

- ✧ Sometimes implemented as part of a "Controlled Franchised" agreement.
  For example, in the apparel industry, Mango let franchisees select styles while Mango controls initial shipment & replenishments, fully liable for excess inventory.



\$b: buy-back price

\$t: Shipping cost (paid by retailer)

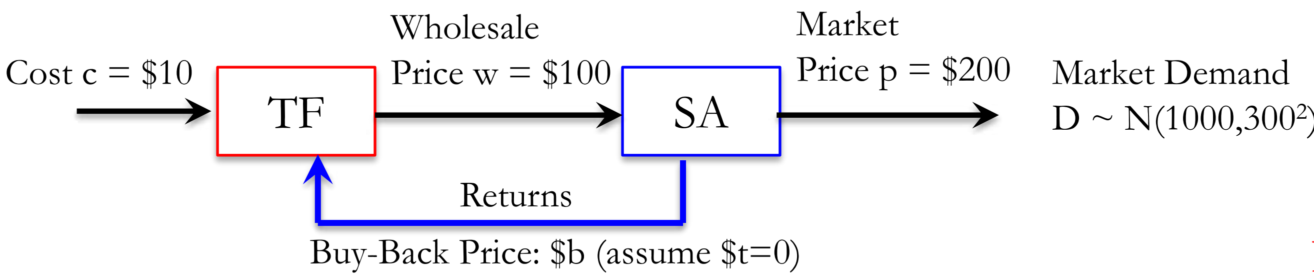
$Cs^R = p-w$ and $Ce^R = w-b+t$

Coordinating buy-back contract: (w,b)

$$\text{Incentive Alignment:} \quad SL^R = SL^C \iff \frac{p-w}{p-b+t} = \frac{p-c}{p-u} \iff b = t + p - (p-w)\,\frac{p-u}{p-c}$$

Example (cont'd)

Tech Fiber (TF) produces jacket and sells to Ski Adventure (SA) which sells them in the market. Unsold jackets have no salvage value. Should TF be willing to buy back unsold jackets? Why? What buy back price should be offered?



No coordination

Buy Back Price $b	Optimal Order Q	Sales Min{D,Q}	Leftover Max{0,Q-D}	Lost Sales Max{0,D-Q}	Expected Profits		
					Retailer	Supplier	Supply Chain
$0.00	1000.0	880.3	119.7	119.7	$76,063	$90,000	$166,063
$10.00	1019.8	890.0	129.8	110.0	$77,310	$90,484	$167,794
$20.00	1041.9	900.1	141.8	99.9	$78,666	$90,936	$169,602
$30.00	1066.9	910.8	156.1	89.2	$80,154	$91,338	$171,492
$40.00	1095.6	922.1	173.5	77.9	$81,799	$91,663	$173,462
$50.00	1129.2	934.0	195.2	66.0	$83,638	$91,868	$175,506
$60.00	1169.8	946.5	223.2	53.5	$85,724	$91,886	$177,610
$70.00	1220.9	959.7	261.2	40.3	$88,136	$91,598	$179,733
$80.00	1290.2	973.4	316.8	26.6	$91,005	$90,776	$181,781
$90.00	1400.6	987.3	413.2	12.7	$94,601	$88,860	$183,461
$99.999	2279.5	1000.0	1279.5	0.0	$99,999	$77,207	$177,205
$94.74	1493.5	993.7	499.7	6.3	$96,743	$87,069	$183,812
$56.00	1152.5	941.4	211.1	58.6	$84,856	$91,907	$176,763

Max supply chain profits

$$b = t + p - (p - w)\frac{p - u}{p - c}$$

$$= 200 - (200 - 100)\frac{200}{200 - 10}$$

$= \$94.74$

(see Excel file: "Newsvendor_Normal.xlsx" available on Canvas)

Module 7: Key Ideas

- Beer Game and Bullwhip Effect
  - ✓ Demand Forecasting Error: Better inventory visibility and aggregate data.
  - ✓ Pricing and Promotions: EDLP
  - ✓ Rationing and Gaming: Data driven inventory allocation

- Supply Chain Coordination.
  - ✓ Information Sharing
  - ✓ Order Smoothing: Reduce by implementing “Heijunka”, 3PLs
  - ✓ Vendor Managed Inventory (JITD)
  - ✓ Myopic vs. Global Optimization: Increase the size of the pie
    - ✓ Contracting to align economic incentives
  - ✓ Strategic change needs top management involvement.

After Class Activities

- GROUP:
  WORK ON REPORTS AND PRESENTATION